

Techniques in Stochastic Analysis

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Abstract

Some informal lectures notes on techniques in stochastic analysis.

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Part I

Markov Semigroup Theory

Motivation for learning this In the previous set of lectures, we briefly mentioned **Langevin dynamics**. Given a target distribution π , you can initialize your samples anywhere X_0 then time-evolve them according to

$$X_{t+h} = X_t + h \nabla \log \pi(X_t) + \sqrt{2h} Z \quad \text{Discrete Time} \quad (0.1)$$

$$dX_t = \nabla \log \pi(X_t) dt + \sqrt{2} dW_t \quad \text{Continuous Time} \quad (0.2)$$

and as $t \rightarrow \infty$, your samples will converge to $X_\infty \sim \pi$ (target measure).

Langevin fell under a special class of dynamics called **Markov Processes / Markov Chains**. These lectures will walk us through some of the tools & techniques to analyze these processes. Markov chains show up in many disciplines, so to give the reader a consistent viewpoint, I'll adopt my POV (which is a computational statistician interested in the math; meaning you only have oracle access to π , and sometimes only have oracle access to the normalized density $\tilde{\pi}$ associated with $\pi = \tilde{\pi}/Z$), but hopefully the lessons can give the reader enough to generalize this knowledge to other problems.

From this view point, there's a couple questions that jump out to me

1. Can we develop other algorithms which can generate samples from π ?
2. Once we specify a sampling algorithm, can we quantify how quickly samples will converge to π ?
3. Many algorithms are easy to specify in continuous time, but we implement only in discrete time. Can we quantify how the discretization affects the analysis?

To keep us grounded, we'll work through these questions by looking back to Langevin, but the theory will stay general.

1 Discrete-Time Markov Chains

Consider a stochastic process $(X_t)_t$, and it's corresponding marginal over time $\text{Law}(X_t) := \pi_t$. The name of the game is to track how a state π_t evolves over time. In these lecture notes, we'll look at a specific class of dynamics, called Markov Chains

Definition 1 (Time-Homogenous Markov Chain). A discrete-time stochastic process $(X_t)_t$ is a Markov Chain if $p(X_{t+1}|X_t, \dots, X_0) = p(X_{t+1}|X_t)$, denoted as the transition kernel. It is time-homogenous because the kernel is time-independent.

In this case, the dynamics of the system are completely governed by the transition kernel.

- When the state space Ω is continuous, π is a probability density function ($\pi \geq 0$ and $\int_{\Omega} \pi(x) dx = 1$). The dynamics are

$$\pi_{t+1}(y) = \int P(x, y) \pi_t(x) dx, \quad P(x, y) = p(X_{t+1} = y | X_t = x) \quad (1.1)$$

The notation of $P(x, y)$ is chosen s.t. it can be read, the probability of starting at x and ending up at y .

- If the state space is discrete, then π_t is a vector in the probability simplex (π_t is a vector, $[\pi_t]_i \geq 0$ and $\sum_i [\pi_t]_i = 1$). The dynamics turn into a matrix equation

$$\pi_{t+1} = \pi_t P, \quad P_{ij} = \mathbb{P}(X_{t+1} = j | X_t = i) \quad (1.2)$$

Because P_{ij} represents probabilities $\sum_j P_{ij} = 1$. This is called a **stochastic matrix**

Some other fun facts,

- When you apply P multiple times, you're evolving multiple steps. E.g. P^k time evolves for k steps.
- The eigenvector $\pi = \pi P$ corresponds to the **stationary distribution** / fixed point.

Example. Looking back at our motivating example, Langevin dynamics. It is also a Markov Chain

$$dX_t = \nabla \log \pi(X_t) dt + \sqrt{2} dW_t \quad (1.3)$$

$$X_{t+h} = X_t + h \nabla \log \pi(X_t) + \sqrt{2h} Z \quad (\text{Discretization}) \quad (1.4)$$

First, let's prove it's Markovian

Now let's derive the transition kernel $p(X_{t+h} | X_t = x)$

$$X_{t+h} = X_t + h \nabla \log \pi(X_t) + \sqrt{2h} Z \quad (1.5)$$

$$X_{t+h} | X_t = x \stackrel{d}{=} x + h \nabla \log \pi(x) + \sqrt{2h} Z \quad (1.6)$$

The probability distribution of Z is $\mathcal{N}(0, \mathbb{I})$. We can use the rules of Gaussian random variables

$$b + \sigma Z \sim \mathcal{N}(b, \sigma^2 \mathbb{I}) \quad (1.7)$$

$$\therefore x + h \nabla \log \pi(x) + \sqrt{2h} Z \sim \mathcal{N}(x + h \nabla \log \pi(x), 2h \mathbb{I}) \quad (1.8)$$

So if the RHS is drawn from a Gaussian, therefore the LHS is also a Gaussian. Therefore the probability distribution of $X_{t+h} = x' | X_t = x$ is

$$p(x' | x) = \frac{1}{Z} \exp \left(-\frac{(x' - (x + h \nabla \log \pi(x)))^2}{4h} \right) \quad (1.9)$$

1.1 Stationary Distribution

Since our objective is to construct a Markov chains which has a stationary distribution. It turns out a chain has a stationary distribution under two sufficient conditions

1. *Irreducibility:* It is possible to get from x to another state x' w/ probability > 0 in a finite number of steps
2. *Aperiodicity:* It is possible to return to any state at any time. I.e. $\exists n$ s.t. $\forall \omega \in \Omega$ and $\forall n' > n$, $\mathbb{P}(X_{n'} = i | X_0 = i) > 0$.

The first prevents absorbing states, e.g. states that we can never leave. The second prevents transition operations such as

$$T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (1.10)$$

which is a chain that alternates between 1 and 2 without ever settling. These conditions constrain the types of Markov chains to get a nice result.

Theorem 1 (Perron-Frobenius). An irreducible & aperiodic Markov Chain P

1. Admits a unique stationary distribution π
2. The stationary distribution π is the eigenvector of P corresponding to the largest eigenvalue $|\lambda| = 1$.

The proof is more about linear algebra than Markov Chains, so it's omitted.

The importance of the theorem is that the eigenvector corresponding to the largest eigenvalue is the stationary distribution. Visualize the coordinate transformation of the matrix P . Repeated applications of P will slowly project your vector along eigenvector

Fact. For an irreducible & aperiodic Markov Chain P with stationary distribution π , repeated applications of P will make any initialization π_0 converge to π .

$$\lim_{k \rightarrow \infty} \pi_0 P^k = \pi \quad (1.11)$$

The proof will be shown in the section on convergence rates.

Detailed Balance and Metropolis-Hastings Algorithm You might wonder, given oracle access to π , how can we construct a Markov chain P which emits π as the stationary distribution.

Well, say you're at the stationary distribution π , then you'd hope for the following properties

1. Invariant under time-evolution: $\pi = \pi P$
2. Invariant under reverse time-evolution $\pi_i = \sum_j p(x_{t+1} = j | x_t = i) \pi_j \implies \pi = P^T \pi$

Definition 2 (Detailed Balance). It is the balance between transition rates

$$\pi(y)p(x|y) = \pi(x)p(y|x) \quad (1.12)$$

where p is the transition kernel of a Markov Chain.

Fact. Let p be the transition kernel associated with a Markov chain. If (p, π) satisfies detailed balance, then π is the stationary distribution of the chain.

Proof.

$$\pi(y)p(x|y) = \pi(x)p(y|x) \quad (1.13)$$

Just integrate w.r.t. y

$$\sum_y \pi(y)p(x|y) = \pi(x) \sum_y p(y|x) \quad (1.14)$$

$$\sum_y \pi(y)p(x|y) = \pi(x) \quad \sum_y p(y|x) = 1 \quad (1.15)$$

$$\pi P = \pi \quad (1.16)$$

and then x

$$\pi(y) \sum_x p(x|y) = \sum_x \pi(x)p(y|x) \quad (1.17)$$

$$\pi = \pi P \quad (1.18)$$

"It seems too easy", Jonathan Goodman. $\square$

As the person running experiments, I simply want to specify a target distribution π & a proposal kernel p , and check which one works well. However not every transition kernel satisfies detailed balance for arbitrary π . Perhaps we can create a framework which always emits detailed balance.

Consider

Metroplst-Hasting rejection is a way to create a transition kernel $p(\cdot|x)$ which satisfies detailed balance.

$$\alpha(x|x') = \min \left(\frac{\pi(x)p(x'|x)}{\pi(x')p(x|x')}, 1 \right) \quad (1.19)$$

Algorithm 1 Metropolis–Hastings

Require: Target π , proposal $p(\cdot | x)$, initial X_0 , steps N

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1: for  $t = 0, 1, \dots, N - 1$  do
2:   Propose  $X' \sim p(\cdot | X_t)$ 
3:    $\alpha \leftarrow \min \left( 1, \frac{\pi(X') p(X_t | X')}{\pi(X_t) p(X' | X_t)} \right)$ 
4:   Draw  $u \sim \text{Uniform}(0, 1)$ 
5:   if  $u \leq \alpha$  then
6:      $X_{t+1} \leftarrow X'$   $\triangleright$  accept
7:   else
8:      $X_{t+1} \leftarrow X_t$   $\triangleright$  reject
9:   end if
10: end for
11: return  $(X_0, \dots, X_N)$ 
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Example (Metropolis-Adjusted Langevin (MALA)). For Langevin dynamics, we proved that it has the stationary distribution π . However this was in continuous time. Does this still hold when we discretized time?

Recall the Euler-Maruyama discretized Langevin

$$X_{t+h} = X_t + h \nabla \log \pi(X_t) + \sqrt{2h} Z \quad (1.20)$$

$$p(x'|x) = \frac{1}{Z} \exp \left(-\frac{(x' - x - h \nabla \log \pi(x))^2}{4h} \right) \quad (1.21)$$

this is known as **Unadjusted Langevin (ULA)**. It's clear the kernel doesn't satisfy detailed balance (for arbitrary π), as there's no π term in the kernel. Although detailed balance is "sufficient, not necessary" that you converge to π , turns out the discretization makes you sample a biased version of π . We can correct for this bias by using the proposal kernel in the Metropolis-Hasting step.

Let's calculate the acceptance ratio for the ULA kernel. This procedure is called **Metropolis-Adjusted Langevin (MALA)**

$$\log \frac{p(x'|x)}{p(x|x')} = \frac{1}{4h} [(x' - x + h \nabla \log \pi(x))^2 - (x - x' + h \nabla \log \pi(x'))^2] \quad (1.22)$$

$$= \frac{h}{4} (\nabla \log \pi(x)^2 - \nabla \log \pi(x')^2) + \frac{1}{2} \langle x' - x, \nabla \log \pi(x) + \nabla \log \pi(x') \rangle \quad (1.23)$$

which you plug into $\alpha = \min \left(\frac{\pi(x)p(x'|x)}{\pi(x')p(x|x')}, 1 \right)$.

You can also run the Langevin

Problem 1. The previous MALA algorithm was stated for single step. Consider a modification to the MALA algorithm for multiple steps

What is the new correction factor $p(x'|x)/p(x|x')$ in the Metropolis acceptance ratio?

Sampling without Detailed Balance For example: sequential monte carlo & diffusion models, bouncy particle sampler.

1.2 Convergence to Stationary Distribution

It's nice that we can construct Markov chains where we know the stationary distribution. However, I need to run this and get some results, so perhaps we can engineer transition kernels with quicker convergence?

Discrete Space & Discrete Time Markov Chains Let's now prove convergence fact from the previous section, the proof will give us a nice insight into convergence rates.

Fact. Consider an irreducible & aperiodic Markov Chain P with stationary distribution π . Repeated applications of P will make any initialization π_0 converge to π .

$$\lim_{k \rightarrow \infty} \pi_0 P^k = \pi \quad (1.24)$$

Proof. For simplicity (and w/o loss of generality), assume P is diagonalizable with eigen-system $\{(\lambda_i, v_i)\}_{i=1}^d$, where $\lambda_1 > \dots > \lambda_d$. By Perron-Frobenius theorem $\lambda_1 = 1$ and

$$v_1 = \pi.$$

Decompose π_0 into a linear combination of the eigensystem

$$\pi_0 P^k = (c_1 \pi + c_2 v_2 \dots + c_d v_d) P^k \quad (1.25)$$

$$= c_1 (1)^k \pi + c_2 \lambda_2^k v_2 + \dots + c_d \lambda_d^k v_d \quad (1.26)$$

$$\lim_{k \rightarrow \infty} \pi_0 P^k = c_1 \pi + \mathcal{O}(\lambda_2^k) = c_1 \pi \quad (1.27)$$

You can kill off λ_2 (and below) because it's smaller than $\lambda_2 = 1$. Because the state still lives in $\mathcal{P}(\Omega)$, it must normalize, therefore $c_1 = 1$.

Note: power iteration can be generically applied to find the eigenvector of the largest eigenvalue. The convergence rate is generically controlled by the **spectral gap** $|\lambda_1| - |\lambda_2|$. $\square$

Notice, for finite k you're converging to the stationary distribution in corrections of λ_2 (second largest eigenvalue). This means spectral properties of the matrix controls the convergence rate of our Markov Chains.

Definition 3 (Spectral Gap). The spectral gap for a matrix P is defined as

$$\gamma = |\lambda_1| - |\lambda_2| \quad (1.28)$$

For irreducible & aperiodic Markov chains, $\lambda_1 = 1$, meaning the convergence is entirely controlled by λ_2 .

Continuous Space & Discrete/Continuous Time Markov Chains When the state space is continuous, the time-evolution P gets promoted to an operator. And you can imagine finding eigenvalues of some operator will be quite difficult. We need to change our tools a little bit!

2 Continuous-Time Markov Chains

When we go to continuous space Markov Chains, it's easy to start with continuous time and use the theory of ODE/PDEs to make statements, then bound the discretization error.

Definition 4 (Markov Semigroup Operator). Given a Markov Process $(X_t)_{t \geq 0}$, tis semigroup $(P_t)_{t \geq 0}$ acts on test functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$(P_t f)(x) \equiv \mathbb{E}[f(X_t) | X_0 = x] \quad (2.1)$$

Why is it called a semigroup? In math a semigroup is when you have closure under composition $g_1 \circ g_2 \in G$, and have an identity $e \circ g = g \circ e \in G$ (s.t. $e \in G$). If you have closure under inverse, then this is promoted to be a group.

Property 1. Let's figure out in what sense this is a semigroup

1. Identity: $P_0 = \text{Id}$.

Proof: $(P_0 f)(x) = \mathbb{E}[f(X_0) | X_0 = x] = f(x)$, so P_0 acts as an identity operator.

2. Composition: $P_t \circ P_s = P_{t+s}$.

Proof:

$$(P_s f)(x) = \mathbb{E}[f(X_s) | X_0 = x] \quad (2.2)$$

$$(P_t P_s f)(x) = \mathbb{E}[\mathbb{E}[f(X_s) | X_0 = X_t] | X_0 = x] \quad (2.3)$$

$$= \mathbb{E}[\mathbb{E}[f(X_{s+t}) | X_t] | X_0 = x] \quad (2.4)$$

$$= \mathbb{E}[f(X_{s+t}) | X_0 = x] = (P_{t+s} f)(x) \quad (2.5)$$

Definition 5 (Generator). The generator $\mathcal{L}$ is defined as

$$\mathcal{L}f := \lim_{h \downarrow 0} \frac{P_h f - f}{h} \quad (2.6)$$

Example. Let's compute the infinitesimal generator for Langevin dynamics. Since we'll take the time-increment to be small, we can start at the Euler-Maruyama discretization.

$$X_h = X_0 + h \nabla \log \pi(X_0) + \sqrt{2h} Z \quad (2.7)$$

where $Z \sim \mathcal{N}(0, \mathbb{I})$. To compute $P_h f = \mathbb{E}[f(X_h) | X_0 = h]$, we can apply a test function f to X_h , and then Taylor expand to second order, that way remaining terms are order h (taking the limit $h \rightarrow 0$ in the infinitesimal generator will kill $h^{3/2}$ terms).

$$f(X_h) = f(X_0) + f'(X_0)(X_h - X_0) + \frac{1}{2} f''(X_0)(X_h - X_0)^2 + \mathcal{O}((X_h - X_0)^3) \quad (2.8)$$

$$\begin{aligned} &= f(X_0) + \langle \nabla f(X_0), -h \nabla \log \pi(X_0) + \sqrt{2h} Z \rangle \\ &\quad + \frac{1}{2} (-h \nabla \log \pi(X_0) + \sqrt{2h} Z)^T \nabla^2 f(X_0) (-h \nabla \log \pi(X_0) + \sqrt{2h} Z) \end{aligned} \quad (2.9)$$

Now we can take the expectation value

$$\mathbb{E}[f(X_h) | X_0 = x] = -h \langle \nabla f(x), \nabla \log \pi(x) \rangle + h \text{Tr}[\nabla^2 f(x)] \quad (2.10)$$

Plugging into the definition of the generator

$$\mathcal{L}f = \langle f, \nabla \log \pi \rangle + \Delta f \quad (2.11)$$

Problem 2. Find the infinitesimal generator associated for a generic stochastic process

$$dX_t = \mu_t(X_t)dt + \sigma_t dW_t \quad (2.12)$$

Does it look familiar?

2.1 Convergence to Stationary Distribution

To quantify the convergence rate to the stationary distribution, we could inspect a "distance" measure between the marginal p_t of the Markov Chain and the target π , and see if its growing / shrinking in time.

Definition 6 (*f*-Divergence). Given a convex function $f : \mathbb{R} \rightarrow \mathbb{R}$, the *f*-Divergence is

$$D_f(p||\pi) \equiv \mathbb{E}_p \left[f \left(\frac{p}{\pi} \right) \right] = \int p(x) f \left(\frac{p(x)}{\pi(x)} \right) dx \quad (2.13)$$

A common choice for $f := \log$; this choice is called the Kullback-Leibler (KL) Divergence

$$\text{KL}(p||\pi) := D_{\log}(p||\pi) = \int p(x) \log \left(\frac{p(x)}{\pi(x)} \right) dx \quad (2.14)$$

It is preferred by computational statisticians because it can be numerically estimated $\text{KL}(p||\pi) = \mathbb{E}_p[\log p] - \mathbb{E}_p[\log \pi]$

Property 2. 1. Not symmetric: $D_f(p||\pi) \neq D_f(\pi||p)$

2. Positive-definite: $D_f(p||\pi) \geq 0$, and $D_f(p||q) = 0 \iff p = q$

I am introducing this new family of divergences, instead of using $\text{TV}(p, \pi) = \frac{1}{2} \int |p(x) - \pi(x)| dx$ because the TV is not differentiable.

Example. Let's compute the *f*-Divergence's time evolution under Langevin. Recall the law for Langevin for a target measure $\pi \propto e^{-U(x)}$ is

$$\partial_t p_t = -\nabla \cdot (\nabla U p_t) + \Delta p_t \quad (2.15)$$

$$\partial_t D_f(p_t||\pi) = \int \dot{p}_t(x) f \left(\frac{p_t(x)}{\pi(x)} \right) + \frac{p_t(x) \dot{p}_t(x)}{\pi(x)} f' \left(\frac{p_t(x)}{\pi(x)} \right) \quad (2.16)$$

$$= \int \dot{p}_t(x) [f(r_t) + r_t f'(r_t)] \quad r_t \equiv p_t/\pi \quad (2.17)$$

$$(2.18)$$

If we specialize to the KL divergence, $f := \log$

Definition 7 (Log-Sobolev Inequality (LSI)). π satisfies a Log-Sobolev Inequality with constant $\lambda > 0$, if

$$\forall p \in \mathcal{P}, \quad \text{KL}(p||\pi) \leq \frac{1}{2\lambda} \text{FI}(p||\pi) \quad (2.19)$$

Fact (Bakery-Emery Criterion). Suppose that $\pi = e^{-U}$ with U β -strongly convex and λ -Lipschitz, that is $\lambda \mathbb{I} \preceq \nabla^2 f \preceq \beta \mathbb{I}$. Then π satisfies LSI with constant λ .

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